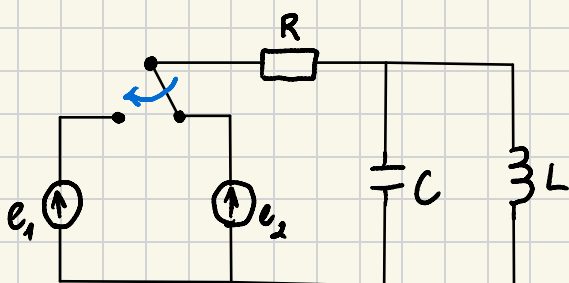


WYKŁAD 12

PRZYKŁAD



$$R = 2 \Omega \quad C = 1 \text{ F} \quad L = 1 \text{ H}$$

$$e_1(t) = 10 \text{ V}$$

$$e_2(t) = 5\sqrt{2} \sin(t + \frac{\pi}{4}) \rightarrow \omega = 1$$

$$E_2 = 5e^{j45^\circ}$$

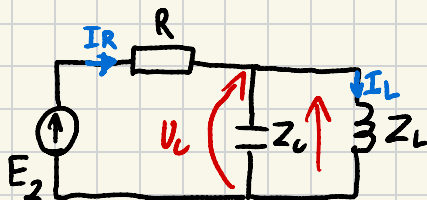
$$Z_L = j\omega L = j$$

$$Z_C = \frac{1}{j\omega C} = -j$$

1° Warunki początkowe (stan ustalony przed przetłoczeniem)

$$i_L(t=0^-) = i_L(0^-)$$

$$u_C(t=0^-) = u_C(0^-)$$



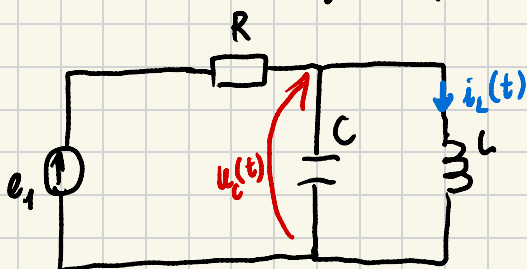
$$Z_{LC} = \frac{Z_L \cdot Z_C}{Z_L + Z_C} = \frac{j \cdot (-j)}{j - j} = \infty$$

$$I_R = 0$$

$$U_C = 5e^{j45^\circ} \rightarrow u_C(t) = 5\sqrt{2} \sin(t + 45^\circ) \rightarrow u_C(t=0^-) = 5\sqrt{2} \sin 45^\circ = 5 \text{ [V]}$$

$$I_L = \frac{U_C}{Z_L} = \frac{5e^{j45^\circ}}{e^{j90^\circ}} = 5e^{-j45^\circ} \rightarrow i_L(t) = 5\sqrt{2} \sin(t - 45^\circ) \rightarrow i_L(t=0^-) = -5 \text{ [A]}$$

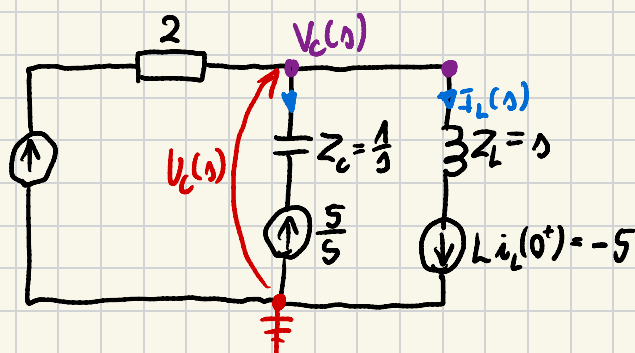
2° Stan nieustalony po przełączeniu



$$i_L(0^+) = i_L(0^-) = -5 \text{ [A]}$$

$$u_c(0^+) = u_c(0^-) = 5 \text{ [V]}$$

Model operatorowy



$$Y \cdot V = I_{zr}$$

$$\left[\frac{1}{2} + \frac{1}{s} + \frac{1}{s} \right] \cdot V_c(s) = \frac{10}{2} + \frac{5}{\frac{1}{s}} + \frac{5}{s}$$

$$\left[\frac{1}{2} + \frac{1}{s} + s \right] \cdot V_c(s) = 5 + \frac{10}{s}$$

$$V_c(s) = \frac{5(1 + \frac{2}{s})}{s + \frac{1}{2} + \frac{1}{s}} = \frac{5s(1 + \frac{2}{s})}{s^2 + \frac{1}{2}s + 1} = \frac{5s + 10}{s^2 + \frac{1}{2}s + 1}$$

$$V_c(t) = \mathcal{L}^{-1} \left[\frac{5s + 10}{s^2 + \frac{1}{2}s + 1} \right] = \frac{5(s + \frac{1}{4}) + \frac{35}{4} \cdot \sqrt{\frac{15}{16}} \cdot \sqrt{\frac{16}{15}}}{(s + \frac{1}{4})^2 + (\sqrt{\frac{15}{16}})^2}$$

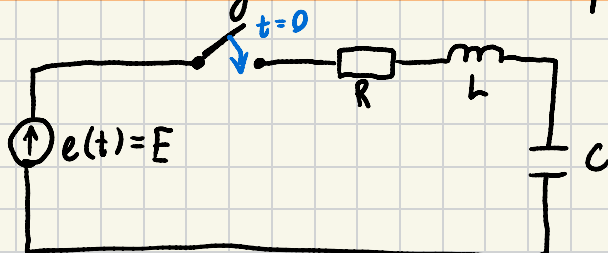
Wynik na następnej stronie

$$\rightarrow u_c(t) = 5e^{-\frac{1}{4}t} \cos\left[\sqrt{\frac{15}{16}}t\right] + \frac{35}{4} \cdot \sqrt{\frac{16}{15}} e^{-\frac{1}{4}t} \sin\left[\sqrt{\frac{15}{16}}t\right]$$

$$1) I_L(s) = \frac{V_c(s) - 5}{s} \rightarrow i_L(t) = \kappa^{-1}[I_L(s)]$$

$$2) i(t) = \frac{10 - u_c(t)}{2} - C \frac{du_c}{dt}$$

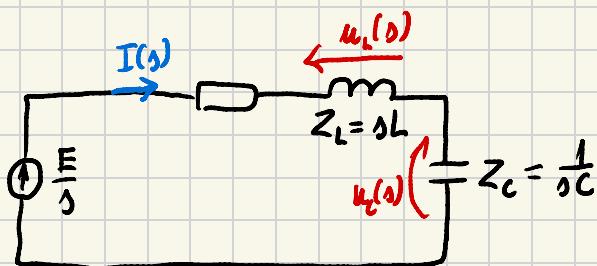
Stan nieustalony w drzewie RLC przy wymuszeniu statycznym



1° Warunki początkowe zerowe

$$i_L(0^-) = 0 \quad u_c(0^-) = 0$$

2° Stan nieustalony po przełączeniu



$$I(s) = \frac{\frac{E}{s}}{R + sL + \frac{1}{sC}} = \frac{E}{s^2L + Rs + \frac{1}{C}} = \frac{\frac{E}{L}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

$$\Delta = \left(\frac{R}{L}\right)^2 - \frac{4}{LC}$$

1) $\Delta > 0$ przypadek aperiodyczny $\rightarrow$ 2 pierwiastki różne
(różne od zera)

$$\left(\frac{R}{L}\right)^2 > \frac{4}{LC} \rightarrow \frac{R}{L} > \frac{2}{\sqrt{LC}} \rightarrow R > 2\sqrt{\frac{L}{C}}$$

2) $\Delta = 0$ przypadek aperiodyczny krytyczny $\rightarrow$ 2 pierwiastki identyczne
 $R = 2\sqrt{\frac{L}{C}}$

3) $\Delta < 0$ przypadek oscylacyjny (periodyczny)
 $R < 2\sqrt{\frac{L}{C}}$

1) PRZYPADEK APERIODYCZNY ($R > 2\sqrt{\frac{L}{C}}$)

$$I(s) = \frac{\frac{E}{L}}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{\frac{E}{L}}{(s-s_1)(s-s_2)}$$

$$\Delta = \left(\frac{R}{L}\right)^2 - \frac{4}{LC} > 0$$

$$\sqrt{\Delta} = \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}$$

$$s_{1/2} = \frac{-\frac{R}{L} \pm \sqrt{\Delta}}{2}$$

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$i(t) = \lim_{s \rightarrow s_1} I(s) e^{st} (s-s_1) + \lim_{s \rightarrow s_2} I(s) e^{st} (s-s_2)$$

CIĄG DALSZY NA NASTĘPNEJ

$$i(t) = \frac{\frac{E}{L} e^{\lambda t} (\lambda - \lambda_1)}{(\lambda - \lambda_1)(\lambda - \lambda_2)} \Big|_{\lambda = \lambda_1} + \frac{\frac{E}{L} e^{\lambda t} (\lambda - \lambda_2)}{(\lambda - \lambda_1)(\lambda - \lambda_2)} \Big|_{\lambda = \lambda_2}$$

$$i(t) = \frac{E}{L(\lambda_1 - \lambda_2)} e^{\lambda_1 t} + \frac{E}{L(\lambda_2 - \lambda_1)} e^{\lambda_2 t}$$

$$i(t) = \frac{E}{L(\lambda_1 - \lambda_2)} [e^{\lambda_1 t} - e^{\lambda_2 t}]$$

$$u_L(t) = L \frac{di}{dt} = \frac{E \cdot L}{L(\lambda_1 - \lambda_2)} [\lambda_1 e^{\lambda_1 t} - \lambda_2 e^{\lambda_2 t}]$$

$$u_c(t) = E - R i(t) - u_L(t)$$

2) PRZYPADEK APERIODYCZNY KRYTYCZNY

$$\Delta = 0$$

$$\gamma_1 = \gamma_2 = -\frac{R}{2L}$$

$$I(s) = \frac{\frac{E}{L}}{(s - \gamma_1)^2} \quad L=2$$

$$i(t) = \lim_{s \rightarrow \gamma_1} \frac{d}{ds} \left[\frac{\frac{E}{L}}{(s - \gamma_1)^2} \cdot e^{st} \cdot (s - \gamma_1)^2 \right] = \frac{d}{ds} \left[\frac{E}{L} \cdot e^{st} \right] = \frac{E}{L} t e^{st}$$

$$u_L(t) = L \frac{di}{dt} = E [e^{\gamma_1 t} + \gamma_1 t \cdot e^{\gamma_1 t}] = E e^{\gamma_1 t} \left[1 - \frac{R}{2L} t \right]$$

$$u_C(t) = E - R i(t) - u_L(t)$$

3) PRZYPADEK OSCYLACYJNY $\Delta < 0$ ($R < 2\sqrt{\frac{L}{C}}$)

$$I(s) = \frac{\frac{E}{L}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

$$\Delta = \left(\frac{R}{2L}\right)^2 - \frac{1}{LC} < 0$$
$$\left(\frac{R}{2L}\right)^2 < \frac{1}{LC}$$

$$I(s) = \frac{\frac{E}{L} \cdot \omega \cdot \frac{1}{\omega}}{\underbrace{\left(s + \frac{R}{2L}\right)^2}_{\alpha} + \underbrace{\left(\frac{1}{LC} - \left(\frac{R}{2L}\right)^2\right)}_{\omega^2}}$$

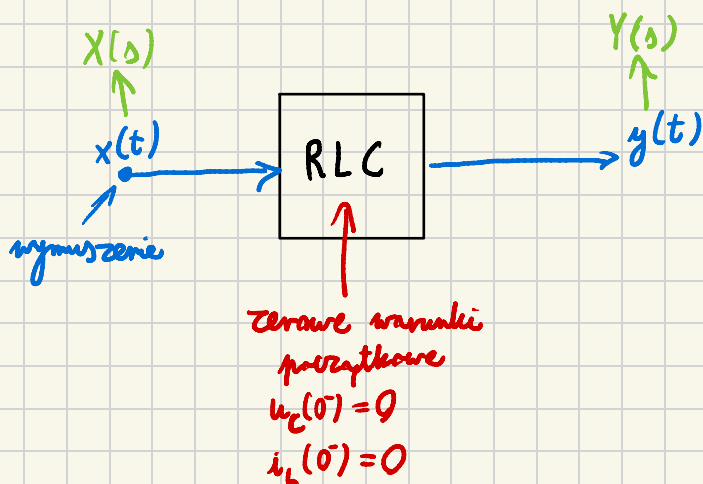
$$\alpha = \frac{R}{2L}, \quad \omega = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

$$i(t) = \frac{E}{\omega L} \cdot e^{-\alpha t} \cdot \sin \omega t$$

$$u_L(t) = L \frac{di}{dt} = \frac{E}{\omega} [-\alpha e^{-\alpha t} \sin \omega t + e^{-\alpha t} \cdot \omega \cdot \cos \omega t]$$

$$u_C(t) = E - R \cdot i(t) - u_L(t)$$

TRANSMITANCJA OPERATOROWA



$$H(s) = \frac{Y(s)}{X(s)} \quad \leftarrow \text{transmitancja operatorowa}$$

Znając $H(s)$ i wymuszenie $X(s)$ można obliczyć

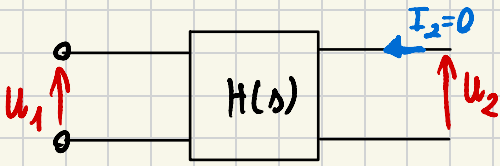
1) $Y(s) = H(s) \cdot X(s)$

2) $y(t) = \mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1}[H(s) \cdot X(s)] \quad \leftarrow \text{stan nieustalony}$

3) jeśli założyc $s = j\omega$ ($\alpha = 0$) $\rightarrow Z_L = sL \Big|_{s=j\omega} = j\omega L \quad \leftarrow \text{stan ustalony przy wymuszeniu o pulsacji } \omega$

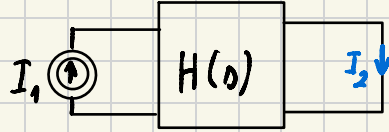
$$Z_C = \frac{1}{sC} \Big|_{s=j\omega} = \frac{1}{j\omega C}$$

1) Transmitancja napięciowa



$$H(s) = \frac{u_2(s)}{u_1(s)} \Big|_{I_2=0}$$

2) Transmitancja prądowa



$$H(s) = \frac{I_2(s)}{I_1(s)} \Big|_{u_2=0}$$